# Corbin J. Allison

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## Education

**University of Colorado Boulder**, Boulder, CO — Expected May 2027
Master of Science in Physics

- Graduate coursework in quantum mechanics, electromagnetism, statistical mechanics, quantum optics, and theoretical and computational chemistry
- Comprehensive exam review paper (in progress) on the thermal Hall effect in the Kitaev magnet $\alpha$-$RuCl_3$, comparing Majorana fermion, magnon, and phonon explanations
- Monte Carlo study of the 2D Ising model in Python, using the Metropolis and Wolff cluster algorithms with finite-size scaling near the critical temperature

**Kansas State University**, Manhattan, KS — May 2024
Bachelor of Science in Physics, Minors in Mathematics and Computer Science

- University Honors Program - Thesis: Avoiding pseudo-resonances in theoretically obtained attosecond-controlled photoionization asymmetries in methyloxirane
- Undergraduate and graduate coursework in mathematics and physics
- James R. Macdonald Memorial Scholarship, 2023 - 2024

## Research Experience

**JILA, University of Colorado Boulder**, Boulder, CO — October 2025 - July 2026
Graduate Researcher | Advisor: Dr. Andreas Becker

- Theoretical study of ionization and nonlinear upconversion of argon and xenon in intense multichromatic laser fields, toward vacuum-ultraviolet excitation of the $^{229m}$Th nuclear clock transition
- Time-dependent Schrödinger equation simulations in the single-active-electron approximation in Fortran 90, with analysis of photoelectron spectra, angular distributions, and high-harmonic yields in Python

**James R. Macdonald Laboratory**, Manhattan, KS — May 2023 - January 2024
Undergraduate Research Assistant | Advisor: Dr. Vinod Kumarappan

- Experimental study of chiral molecules with the RABBITT technique (reconstruction of attosecond beating by interference of two-photon transitions)
- High vacuum systems, velocity map imaging (VMI) and microchannel plate (MCP) detection, and LabVIEW data acquisition

**Kansas State University Department of Physics**, Manhattan, KS — June 2022 - August 2024
Undergraduate Research Assistant and NSF REU Participant | Advisor: Dr. Loren Greenman

- Theoretical study of attosecond control of ionization asymmetries in chiral molecules using time-dependent perturbation theory, the basis of the Bachelor's thesis
- Hartree-Fock calculations in Molpro and photoionization calculations in ePolyScat on the Beocat cluster and NERSC Perlmutter, using Fortran 90, Bash, and Python
- Collaboration with groups at ETH Zürich and Freie Universität Berlin, leading to the publications below

## Publications

Han, M., Ji, J.-B., Blech, A., Goetz, R. E., **Allison, C.**, Greenman, L., Koch, C. P., & Wörner, H. J. (2025). Attosecond control and measurement of chiral photoionization dynamics. *Nature*, 645, 95–100. https://doi.org/10.1038/s41586-025-09455-4

Goetz, R. E., Blech, A., **Allison, C.**, Koch, C. P., & Greenman, L. (2025). Continuum-electron interferometry for enhancement of photoelectron circular dichroism and measurement of bound, free, and mixed contributions to chiral response. *Physical Review Research*, 7, L032036. https://doi.org/10.1103/8f7h-6nfc

## Presentations

APS Division of Atomic, Molecular and Optical Physics (DAMOP), Providence, RI, June 2026. **C. Allison**, K. Finger, and A. Becker. "Ionization and Nonlinear Upconversion of Argon in Intense Multichromatic Fields" (poster).

Gordon Research Conference on Quantum Control of Light and Matter, Newport, RI, August 2023. **C. Allison**, R. E. Goetz, A. Blech, C. P. Koch, and L. Greenman. "Control of Photoelectron Circular Dichroism with RABBITT" (poster).

## Professional Experience

**TollWarden, LLC** (tollwarden.com) July 2026 - Present
Founder and Software Engineer

- Payment security service for AI agents paying with x402 stablecoin micropayments, returning an Ed25519-signed allow, flag, or block verdict bound to a hash of the exact payment
- Detection of prompt-injection-triggered payments, replay, address poisoning, overpayment, and secret leakage, measured against an evaluation corpus of attack and benign cases
- TypeScript server with 763 automated tests, published on npm and the MCP Registry, with TypeScript and Python SDKs and integrations for LangChain, CrewAI, the Vercel AI SDK, NVIDIA NeMo, and Coinbase AgentKit

**Heartland Electric, Inc.**, Fort Riley, KS May 2015 - August 2022
Electrician

- Industrial, commercial, and residential installation, troubleshooting, and repair, including independent work on SCADA, variable frequency drive, and motor control systems

## Teaching Experience

**Atomic and Molecular Spectra**, University of Colorado Boulder August 2026 - Present
Graduate teaching assistant, grading for a graduate course

**Electronics for the Physical Sciences**, University of Colorado Boulder August 2025 - December 2025
Graduate teaching assistant, managing lab sections, diagnosing circuit problems, and grading homework

**General Physics 1**, University of Colorado Boulder August 2024 - Present
Graduate teaching assistant, leading recitations, grading homework, and holding office hours

**Descriptive Physics and Concepts of Physics**, Kansas State University August 2021 - May 2023
Undergraduate teaching assistant and lab coordinator, leading lab sections, grading lab reports and exams, and training other teaching assistants

## Skills

**Programming languages:** Python, TypeScript/JavaScript, Fortran, C/C++/C#, Java, Bash, LaTeX

**Tools:** Node.js, Git, GitHub Actions, Docker, Jupyter, Molpro, LabVIEW, MATLAB, Mathematica